

# Taha Disbudak

Pittsburgh, PA | mtdi2024@mymail.pomona.edu | [website](#) | Domestic Student

## EDUCATION

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### Pomona College

Computer Science, GPA 3.95/4.0

Relevant Coursework: Data Structures and Algorithms, Randomized Algorithms, Calculus III, Linear Algebra, Data Science with R

Claremont, CA

May 2028

## EXPERIENCE

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### Data Engineering Intern

TD Bank

Mt Laurel, NJ

May 2026- August 2026

- Engineered an LLM-powered data extraction agent in Python to automate the file transformation between Azure and in house data storage. Parsing complex .xlsx files into token-optimized JSON to minimize API costs and enhance AI context window efficiency.
- Spearheaded the refactoring of enterprise-level PySpark and Python data pipelines for cross-selling and FX exposure modules, engineering automated unit testing frameworks to validate core business logic and improve code maintainability.
- Orchestrated technical infrastructure provisioning for a new Power BI reporting initiative, collaborating across Agile teams to configure Azure cloud environments, Active Directory security groups, and GitHub enterprise repositories while securing architectural approvals from senior engineering executives and Data Technology Leads.

### Computer Science Research Intern | Co-author [\[poster\]](#)[\[link to publication\]](#)

Human-Computer Interaction Lab, Pomona College & UC Irvine Collaboration:

Dr. Alexandra Papoutsaki (Pomona), Dr. Daniel Epstein (UC Irvine)

Claremont, CA

May 2025 - August 2025

- Engineered a data engineering pipeline in Python to parse, clean, and analyze 71 anonymized datasets from the Huckleberry baby-tracking app, processing over 230,000 individual log entries.
- Developed and generated publication-ready data visualization time series models using Matplotlib to illustrate key parental tracking behaviors, including adoption, use, and abandonment patterns over a 52-week period.
- Authored modular scripts to ensure research reproducibility, enabling dynamic analysis across multiple tracking categories such as Sleep, Milk Feeds, and Diapers.

### AI Project Lead | [\[github\]](#)

Harvey Mudd & Pomona College P-AI Club

Claremont, CA

September 2025 - December 2025

- Directed Program & Project Management for the TagSafe team of 5 members, building an Artificial Intelligence (AI) LLM-powered tool that generates trademark-safe marketing material for e-commerce listings.
- Developed the core API (Python/FastAPI) for automated United States Patent and Trademark Office data ingestion, featuring Nice class filtering and live/dead status monitoring.
- Drove prompt engineering for keyword extraction, category classification, and model interpretability.
- Conducted interviews for hiring, enforced privacy-first design, and implemented legal disclaimer guardrails.

### Software Engineering Intern

MEGA Insulation Solutions

Elazığ, Türkiye

Summer 2022

- Developed and deployed a genetic algorithm in Java to optimize the company's daily production schedule, which was adopted for use by the production department manager.
- Implemented a selection model using a roulette system and random mutations over 6,000 iterations to generate the most efficient production sequence.

## PROJECT EXPERIENCE

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### ASA DataFest 2025: Real Estate Market Analysis | R, Python

- Applied advanced analytics to a proprietary dataset of over 194,000 commercial lease transactions for Savills, integrating U.S. economic data to develop actionable insights on post-pandemic regional market trends.

### Information Technologies Chatbot | Python, LLM Embedding

- Developed a Retrieval-Augmented Generation (RAG) chatbot with Dr. Aashita Kesarwani (Harvey Mudd), implementing vector embeddings and similarity comparison on web-scraped IT data to answer questions related to Harvey Mudd Information Systems.
- Utilized supercomputers via shell access for high-performance computing and applied prompt engineering to test for safety, optimize accuracy, and ensure user-friendly interactions.

## SKILLS

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**Technical Skills:** GitHub, Genetic Algorithms, Statistical Analysis, Data Cleaning, Data Engineering, Data Visualization, Advanced Analytics, Process Automation, Application Development, Prompt Engineering, LLMs, Data Structures

**Languages and Libraries:** Python, Java, R, C, FastAPI, Pandas, Matplotlib, Haskell